

Корни из единицы и комплексная экспонента

Домашнее задание

1 сентября 2026 года

Теоретический минимум. Roots of unity and the filter $\frac{1}{n} \sum_k \varepsilon^{-km}$; cyclotomic polynomials; the real subfield $\mathbb{Q}(\cos \frac{2\pi}{n})$ of degree $\varphi(n)/2$; de Moivre's formula and $\cos n\theta$ as a polynomial in $\cos \theta$; Gauss sums; the identity $\prod_{k=1}^{n-1} 2 \sin \frac{k\pi}{n} = n$.

1. (Gauss) For an odd prime p let $G = \sum_{k=0}^{p-1} e^{2\pi i k^2/p}$. Prove that $|G| = \sqrt{p}$, and determine G^2 in terms of the quadratic residue symbol.
2. (MIPT 2018) Do there exist rational numbers a, b, c, d with

$$\sqrt[3]{2} = a + b \cos \frac{2\pi}{7} + c \cos \frac{4\pi}{7} + d \cos \frac{6\pi}{7} ?$$

Answer the same question for $\sqrt[5]{2}$ in place of $\sqrt[3]{2}$, and explain where the two answers come from.

3. Prove that $\cos n\theta = T_n(\cos \theta)$ for a polynomial $T_n \in \mathbb{Z}[x]$ with leading coefficient 2^{n-1} , and deduce that $2 \cos \frac{2\pi}{n}$ is an algebraic integer of degree $\varphi(n)/2$.
4. (IMC) Determine whether there exist integers $m_1, \dots, m_{15}$ such that $\sum_{k=1}^{15} m_k \arctan k = \arctan 16$. (*Подсказка: гауссовы целые и однозначность разложения на простые.*)
5. Prove that $\sum_{k=0}^{n-1} |1 - \varepsilon^k| = 2 \cot \frac{\pi}{2n}$ and interpret the answer as the perimeter of a star polygon.
6. Prove that a complex polynomial with $|p(z)| = 1$ for all $|z| = 1$ is of the form cz^n with $|c| = 1$; deduce that a trigonometric polynomial of modulus 1 everywhere is $ce^{in\theta}$.

7. Evaluate $\sum_{k=1}^{n-1} \varepsilon^k / (1 - \varepsilon^k)$ and $\sum_{k=1}^{n-1} \frac{1}{1 - \varepsilon^k}$, and deduce the value of $\sum_{k=1}^{n-1} \cot \frac{k\pi}{n}$.
8. Prove that the cyclotomic polynomial $\Phi_n(z) = \prod_{(k,n)=1} (z - \varepsilon^k)$ has integer coefficients, compute Φ_p , Φ_{2p} and Φ_{p^2} for a prime p , and deduce that $\prod_{d|n} \Phi_d(z) = z^n - 1$.
9. Prove the Poisson kernel identity $\sum_{k=-\infty}^{\infty} r^{|k|} e^{ik\theta} = \frac{1-r^2}{1-2r \cos \theta + r^2}$ for $0 < r < 1$, and deduce $\sum_{k \geq 1} \frac{\cos k\theta}{k} = -\ln\left(2 \sin \frac{\theta}{2}\right)$ on $(0, 2\pi)$.
10. Prove that the vertices of a regular n -gon $z_k = \varepsilon^k$ satisfy $\sum_k |z - z_k|^2 = n(|z|^2 + 1)$ for every complex z , and find $\max_{|z|=1} \prod_k |z - z_k|$.